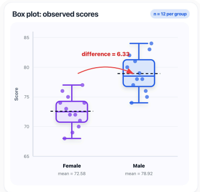
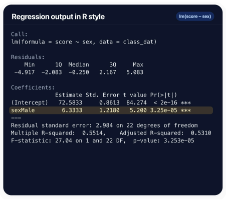
Blooket | Question Set - Answer Key

Name ______________________
Date ______________________
Class ______________________

RSR Week 07: Binary & Polytomous Catergorical Predictors

1. We can easily calculate the difference in mean y between two groups. What doe group) add?





a) All of the above. ✔

b) None of the above

c) A significance test for the difference (a p-value).

d) A predicted y for each group, in a framework that scales to more predictors.

2. The R output models hospital length of stay (days) based on whether the admiss emergency or not. Non-emergencies are the reference. What does the emergency c 2.64 mean? (referring to image)

```
> summary(lm(length_of_stay ~ emergency, data = admissions))

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.220      0.180   23.44  < 2e-16 ***
emergency      2.640      0.290    9.10  1.8e-17 ***
```

a) Emergency admissions stay 2.64 days in the hospital on average.

b) Non-emergency admissions stay 2.64 days longer than emergency admissions.

c) Emergency admissions stay 2.64 days longer than non-emergency admissions on average. ✔

3. You re-fit the same lm() with a different reference category. Which of these stays same?

a) The predicted mean for each group. ✔

b) The signs of the slope coefficients.

c) The intercept value.

4. The output shown fits lm(ldl ~ exercise) on 300 adults. The aim is to predict LDL based on whether or not the person exercises. With 'no' as the reference group. Wl about the intercept (130.00) is correct?

```
# n = 300 adults; exercise is a factor with reference "no"
> mod <- lm(ldl ~ exercise, data = adults)
> summary(mod)

Coefficients:
            Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)   130.00        1.47    88.44   < 2e-16 ***
exerciseyes   -12.00        2.08    -5.77   1.9e-08 ***
```

a) 130 is the average LDL for non-exercisers (the reference group). ✔

b) 130 is the average LDL for exercisers.

c) 130 is the increase in LDL for exercisers compared to non-exercisers.

5. Consider again the regression output table. What does the t value of -5.77 tell u

```
# n = 300 adults; exercise is a factor with reference "no"
> mod <- lm(ldl ~ exercise, data = adults)
> summary(mod)

Coefficients:
            Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)   130.00        1.47    88.44   < 2e-16 ***
exerciseyes   -12.00        2.08    -5.77   1.9e-08 ***
```

a) The estimate is about 5.77 SEs away from zero ✔

b) The estimate is 5.77 mg/dL below the intercept

c) Exercisers have LDL about 5.77 times lower than non-exercisers

6. This model fits lm(sbp ~ activity) on 400 adults, with exercise as the predictor. W mean systolic BP for the moderate group? (referring to image)

```
# n = 400 adults; activity is a factor: sedentary (ref), moderate, vigorous
> mod2 <- lm(sbp ~ activity, data = cohort)
> summary(mod2)

Coefficients:
                 Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)        138.00        1.22   112.98   < 2e-16 ***
activitymoderate    -8.00        1.71    -4.68   3.9e-06 ***
activityvigorous   -16.00        2.15    -7.44   5.4e-13 ***
```

a) 130.00 mmHg ✔

b) 122.00 mmHg

c) 138.00 mmHg